

Quantifying Client Dependence in a Federated Water Distribution Network

Interventional sensitivity analysis, hydraulic unmixing, and observability-driven sensor placement
FedWater — handoff note

Result. Cross-district dependence on the Graeme network is now measured by intervention rather than inferred from correlation. Roughly 78% of the similarity previously observed between districts is hydraulic mixing, not client behaviour. Inverting an interventionally-measured Jacobian removes that component and recovers true district-demand similarity at rank correlation 0.955 (MAE 0.017), retaining 93–100% of the sector structure present in the underlying demand.

Why it matters. A long sequence of negative results — sector tests, drift-attribution, Granger direction — is now explained rather than merely reported. They failed for one reason: correlation is not preserved under linear mixing, so no choice of *which* mixed signal to observe could have worked.

1. Headline numbers

Quantity	Value	Evidence
Hydraulic share of observed similarity	78% (median, demand-weighted)	Jacobian prediction vs simulated independent demands, agree to 0.01
Recovery of true demand similarity	rank corr 0.955 ± 0.040 ; MAE 0.017	20 worlds, paired $p < 0.0001$ vs raw sensors
Sector structure retained after unmixing	93–100% of the true gap	5 consumption maps, recovered $p = 5.0e-06$
Directed supply chain	$A \rightarrow B \rightarrow C \rightarrow D \leftrightarrow E$	Flow transit; District A carries 1.00 of every district's demand
Pressure coupling	within/between ratio 1.1185	Invariant across a 20× operating-point change
Placement criterion	$\sigma_{\min}$ of the district Jacobian	Predicts recovery level and noise decay

2. What was actually built

The final method has four stages, each validated against a case whose answer is known by construction before being applied to Graeme.

- **Flat baseline.** All demands constant and pattern-free, $\text{duration}=0$, one steady-state solve. EPANET's simulation is quasi-static, so this is a Jacobian, not a dynamic response; a longer run would let tanks cycle and reintroduce the drift the design removes.
- **Perturbation sweep.** Each of 113 junctions perturbed individually at three additive levels (339 solves). Additive, not multiplicative, so magnitudes are comparable across nodes and zero-demand junctions are still probed.
- **District Jacobian and unmixing.** Node-level inversion is hopeless (15 sensors, 113 nodes), but district aggregation makes it overdetermined (15×5): $d = J^{\dagger}y$ is well posed. Dependence statistics are then computed on the demand *estimate*, not on the mixed observation.
- **Placement by conditioning.** Unmixing error scales as $1/\sigma_{\min}$, making optimal experiment design the natural placement objective.

Physics checks (all pass)

Check	Result	What it catches
Mass balance	4.85e-06 rel. error	Sign and bookkeeping errors; caught a real one (see §3)

Diagonal dominance	1.000 (weak, 1% tol.)	Ties are physical — a node below the source moves all downstream equally
Reciprocity	7.4e-04 asymmetry	Confirms pressure is symmetric $\Rightarrow$ can never yield direction
Linearity across levels	p05–p95 ratio 0.98–1.05	Bounds how far the linear map extrapolates
Baseline flatness	flow CV 3.7e-07	Tank cycling would reintroduce the removed confound

3. The path, including what failed

The failures are load-bearing: each one eliminated a hypothesis and narrowed the design. They are recorded here because the final method is only defensible in light of what it replaced.

Dead ends that were correctly abandoned

Attempt	Outcome	Why it failed
Granger causality on district signals	Chance-level (1/10 vs physics)	Peak cross-correlation lag = 0 for every pair at every resolution. Quasi-static hydraulics has no transport lag; asymmetries tracked self-predictability (time reversal survived 93%)
Finer time resolution (1h → 15min → 7.5min)	No change; 8× data, zero gain	EPANET solves independent steady states — refining the step gives more steady states, not a lag
Pre-drift month windows	AR R ² rose 0.988 → 0.994	Residual autocorrelation is intrinsic (smooth dynamics), not non-stationarity
Sector tests on residual similarity	p = 0.41; direction inconsistent	Measuring mixed observations; also per-sensor deseasonalization removes the sector profile shape
Decoupled placement (minimise mixing)	Worst raw performer (0.025)	Wrong hypothesis: mixing is not the enemy, unobservability is. Removing mixing also removed signal

Bugs found by the checks, not by inspection

- **Demand multiplier.** Mass balance failed at exactly 0.800 for every node and every level. Graeme .inp ships EPANET's global multiplier at 0.2, so a fifth of each perturbation was applied. Structure was unaffected (a global scalar divides out of every ratio) but magnitudes were understated ~20×, and the Jacobian had been measured at the wrong operating point.
- **Internal-only transit.** District A first appeared hydraulically isolated. It is fed at its own boundary node (pipe 184 → node 110), so its water never enters an internal pipe. Counting edge pipes revealed A as the trunk that carries 1.00 of every district's demand.
- **Stale estimates.** A sector test returned only three distinct per-world values, repeated exactly — the signature of a demand estimate computed once and regrouped, rather than rebuilt per world. Rebuilding it turned a null (p = 0.45) into the headline result (p = 5.0e-06).

4. Two structural findings worth stating on their own

Pressure cannot give direction — this is a theorem. Pressure sensitivity is symmetric by Maxwell–Betti reciprocity (measured asymmetry 7.4e-04 on a looped, heterogeneous network). It therefore measures coupling strength only. Direction must come from flow, which is a routing matrix and manifestly asymmetric. On this single-source network pressure is also nearly non-discriminative: any perturbation moves ~99% of all nodes, and the within/between ratio is 1.12.

The cheap heuristic is near-optimal. d_optimal achieved the best conditioning ($\sigma_{\min}$ 0.365 vs 0.337) but tied variance in recovery (0.950 vs 0.967). variance is purely local, needs no Jacobian and no coordination — which is the right recommendation for a federated deployment.

5. Next steps, in dependency order

#	Step	Why now	Cost
1	Bridge to the FL pipeline: unmix as a preprocessing step before fl_preprocessing	Unmixed demand is locally computable by each client from its own sensors plus one Jacobian column. Tests whether federated artifacts recover sector structure that unmixing makes visible at 0.95	Medium
2	Seed replication: ~10 worlds varying only sim_seed	Every result so far fixes seed 42. This measures the noise floor directly — the one quantity none of the tests had	Low
3	Pre-drift vs post-drift split of recovery	The Jacobian is linear around a flat baseline while worlds drift. Recovery survived at 0.955; this quantifies the margin	Low
4	Real measurement-noise model instead of synthetic fractions	Robustness was tested at 1–10% of sensor SD; params["noise"] is the deployment-relevant case	Low
5	Multi-source / tank network	Mixing becomes state-dependent and a single static J may not suffice. This is the main external-validity limit	High
6	Shape-based regime statistic alongside correlation	Sectors differ in profile shape; correlation captures co-movement. A normalized-weekly-profile distance may be the sharper instrument	Medium

6. Limits to state plainly

- One network, one operating point, static. The Jacobian is valid near the baseline it was measured at; nonlinearity is bounded (p05–p95 ratio 0.98–1.05) but not zero.
- Graeme has a single reservoir and no tanks. Both the pressure-uniformity finding and the sufficiency of a single static Jacobian may not transfer.
- node_weights assumes district demand moves proportionally to node size; BFS-concentrated drift violates this. Recovery held anyway, which is evidence of robustness rather than a reason to ignore it.
- All worlds fix sim_seed = 42; replication so far is across scenarios, not stochastic draws.
- Income regime was untestable — every consumption map in the sweep is all-low income, so that axis had no variation.

7. Artifacts

File	Contents
sensitivity_probe.py	Flat baseline, perturbation sweep, physics checks, influence maps, supply dependency, similarity attribution, unmixing, placement selectors (decoupled, D-optimal), ipywidgets explorer
influence_probe.ipynb	The interventional study end to end: §1–8 probe and maps, §9–11 attribution, §12–14 placement, unmixing and noise robustness
placement_poc.py / dependence_and_placement.ipynb	Observational side: placement strategies, dependence battery, horizon sensitivity, directional diagnostics — the work the probe explains

Every claim above is anchored to a number produced in the notebooks, and every method was validated against a construction with a known answer before being applied to Graeme. Where a hypothesis was wrong — decoupling, finer resolution, drift-window purity — it is recorded as wrong rather than quietly dropped.